

```
package com.mycompany.codealpha_java_student_grade_tracker;
```

```
public class Student {  
    private String name;  
    private double grade;  
  
    public Student(String name, double grade) {  
        this.name = name;  
        this.grade = grade;  
    }
```

```
    public String getName() {  
        return name;  
    }
```

```
    public double getGrade() {  
        return grade;  
    }
```

```
    public String getStatus(){  
        if(grade>=85)  
            return "Excellent";  
        else if(grade>=70)  
            return "Good";  
        else if(grade>=50)  
            return "Pass";
```

```
        else  
            return "Fail";  
    }  
  
}
```

```
package com.mycompany.codealpha_java_student_grade_tracker;
```

```
import java.util.ArrayList;
```

```
import java.util.Scanner;
```

```
public class CodeAlpha_Java_Student_Grade_Tracker {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc=new Scanner(System.in);
```

```
        while (true) {
```

```
            System.out.println("===== STUDENT GRADE TRACKER =====");
```

```
            System.out.println("1. Add Student");
```

```
            System.out.println("2. Display Students");
```

```
            System.out.println("3. Show Statistics");
```

```
System.out.println("4. Search Student");
```

```
System.out.println("5. Exit");
```

```
System.out.print("Enter your choice: ");
```

```
int choice = sc.nextInt();
```

```
sc.nextLine();
```

```
switch (choice) {
```

```
    case 1:
```

```
        addStudent(sc);
```

```
        break;
```

```
    case 2:
```

```
        displayStudents();
```

```
        break;
```

```
    case 3:
```

```
        calculateStatistics();
```

```
        break;
```

```
    case 4:
```

```
        searchStudent(sc);
```

```
        break;
```

```
    case 5:
```

```
        System.out.println("Exiting program...");
        sc.close();
        return;

    default:
        System.out.println("Invalid choice. Try again.");
    }
}
}
```

```
static ArrayList<Student> students = new ArrayList<>();
```

```
public static void addStudent(Scanner sc) {
    System.out.print("Enter student name: ");
    String name = sc.nextLine();

    System.out.print("Enter student grade: ");
    double grade = sc.nextDouble();
    sc.nextLine();

    students.add(new Student(name, grade));

    System.out.println("Student added successfully.");
}
```

```

public static void displayStudents() {

    if (students.isEmpty()) {

        System.out.println("No student data available.");

        return;

    }

    System.out.println("===== STUDENT REPORT =====");

    for (Student s : students) {

        System.out.println(

            "Name: " + s.getName() +

            " | Grade: " + s.getGrade() +

            " | Status: " + s.getStatus());

    }

    System.out.println("=====");

}

```

```

public static void calculateStatistics() {

    if (students.isEmpty()) {

        System.out.println("No student data available.");

        return;

    }

```

```
double total = 0;
```

```
double highest = students.get(0).getGrade();
```

```
double lowest = students.get(0).getGrade();
```

```
String highestStudent = students.get(0).getName();
```

```
String lowestStudent = students.get(0).getName();
```

```
for (Student s : students) {
```

```
    total += s.getGrade();
```

```
    if (s.getGrade() > highest) {
```

```
        highest = s.getGrade();
```

```
        highestStudent = s.getName();
```

```
    }
```

```
    if (s.getGrade() < lowest) {
```

```
        lowest = s.getGrade();
```

```
        lowestStudent = s.getName();
```

```
    }
```

```
}
```

```
double average = total / students.size();
```

```
System.out.println("===== STATISTICS =====");
```

```

        System.out.println("Average Grade : " + average);
        System.out.println("Highest Grade : " + highest + " (" + highestStudent + ")");
        System.out.println("Lowest Grade : " + lowest + " (" + lowestStudent + ")");
        System.out.println("=====");
    }

```

```

public static void searchStudent(Scanner sc) {

    if (students.isEmpty()) {
        System.out.println("No student data available.");
        return;
    }

```

```

        System.out.print("Enter student name to search: ");
        String searchName = sc.nextLine();

```

```

        boolean found = false;

```

```

        for (Student s : students) {
            if (s.getName().equalsIgnoreCase(searchName)) {

                System.out.println("Student Found:");
                System.out.println("Name   : " + s.getName());
                System.out.println("Grade : " + s.getGrade());
                System.out.println("Status : " + s.getStatus() + "");
            }
        }
    }
}

```

```
        found = true;
        break;
    }
}

if (!found) {
    System.out.println("Student not found.");
}
}

}
```